

Instructor: Date:

Learner: Regional Branch:

Program: Center:

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Let's Start!

How can data communicate?



Learning Outcome:

Describe common activities at work to a visitor using short sentences in familiar spoken and written situations.
Understand information about data and its structure in simple spoken and written situations.

Can Do Statements:

- I can identify features related to the applications of data communications.
- I can explain different applications related to data communications.
- I can talk about applications of data communications broadly.

Ask your partner:

- How do you think data can be communicated?
- What do you think are the uses of data communication?





Let's Explore!

Talking about applications of data communications!



Instructions: Read the text below.

- Sasha:** I don't know much about applications of data communications. What do you know about it?
- Peter:** The applications of data communications are the core for networks and the basis for the proper functioning of service providing servers.
- Sasha:** What kind of services do they provide?
- Peter:** Services like World Wide Web pages, electronic mail, online databases, careware and multimedia content.
- Sasha:** Multimedia content like audio and video?
- Peter:** Exactly! These applications are the global bridge between computers and communication. It isn't possible to work without data communications even when casting variables. The demand for data communications is growing every day, and services like DSL and optic cable modems contribute to the growth of these infrastructures.
- Sasha:** Besides, in this new century, applications make lives easier and make communication faster and more efficient by taking advantage of the demand for more electronic methods of communications or by using strategies of evaluation like "call-by-reference" or "call-by-value" in computers programming.
- Peter:** You're right! In general, applications have three specific characteristics. One, delivery; data should be delivered to the correct destination. Two, accuracy; nowadays, data isn't being very accurate and sometimes it is altered in its technical transmission. Three, timeliness; it means as fast as possible. It is not useful to get data delivered late.
- Sasha:** Is there a device that helps with data communications?
- Peter:** Yes, the modem, which is an electronic device that encodes digital information. It produces a signal to reproduce the original digital data. There is also the computer network; it shares different resources and is an interconnection of various computer systems that aren't in the same local place.
- Sasha:** What about teleconferencing? It isn't very popular, right?
- Peter:** It is! This global application refers to meetings that involve people who aren't physically at the same place. Other applications are ethernet cables and wireless computing.





Let's Practice!

False Facts About Applications of Data Communication

Instructions: Use the words from the Word Bank to complete each sentence.

Word Bank

don't / doesn't / aren't / is not / do / it is

1. The applications of data communications _____ represent the core for networks.
2. World Wide Web pages, electronic mails, online databases, care ware and multimedia content _____ services that applications of data communications provide.
3. The demand for data communications _____ growing as we thought.



Let's Practice!

Applications Related to Data

Instructions: Select the correct answer.

1. People think a modem is not an electronic device that _____ the digital information.
Hint: simple present
a. encode
b. encoding
c. encodes
2. Does _____ represent a kind of application that benefits charity?
Hint: Software that benefits charity.
a. careware
b. charity software
c. casting Variable
3. In computer programming, _____ is _____ changes within online databases.
Hint: refers to an evaluation strategy / simple present in negative form.
a. call by careware / isn't make
b. call by value / not making
c. call by reference / doesn't make
4. Computer network _____ different resources and interconnects different systems with _____.
Hint: simple present / different variables.
a. share / careware
b. shares / casting variable
c. sharing / callback
5. _____ is _____ every single electronic application to make communications easier.
Hint: evaluation strategy / present continuous.
a. Call by reference / evaluating
b. Call by reference / evaluates
c. Referring call / evaluating



Extension Activity

Data communication applications.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for specific information of two common applications of data communications you know.
- Make a poster describing some features about them.
- Use programming words to compare them to other applications.
- Formulate some questions to ask for different application related to data communications.
- Share the poster with your partners or instructor.

Self-Assessment

Yes No

- I look for information about common applications of data communications.
- I write important features about applications.
- I compare applications of data communications.
- I answer questions about applications of data communications.

☐	☐
☐	☐
☐	☐
☐	☐



Target Words

- **server** (noun): a computer or program that gives data to other devices
e.g. The applications of data communications are the core of networks and the basis for the proper functioning of service providing servers.
- **multimedia** (adjective): that involves the use of a combination of media
e.g. Servers like World Wide Web pages, electronic mail, online databases, and multimedia content.
- **demand** (noun): a specific requirement by customers
e.g. The demand for data communications is growing every day, and services like DSL and cable modems help the growth of these infrastructures.
- **modem** (noun): an electronic device that receives and sends data
e.g. The demand for data communications is growing every day, and services like DSL and cable modems help the growth of these infrastructures.
- **delivery** (noun): the act of transferring information
e.g. One, delivery. Data should be delivered to the correct destination.
- **accurate** (adjective): when something is precise and with no error
e.g. Nowadays data isn't being very accurate and sometimes it's altered in its transmission.



Learning Points

Adjectives Used in Applications of Data Communications

Adjectives are words that describe the noun of a sentence. There are certain words in the software world to describe **application of data communications**.

These are some adjectives used in applications of data communications with their meaning:

- Electronic: something related to the transmission of information by electronic systems.
- Optic: related to the use of fibers to transmit light or information.
- Actual: that exists in real life.
- Local: that takes part of a particular and delimited area.
- Global: that is worldwide or involves the whole world.
- Technical: that involves science or industry.

Examples

- *These applications are the global bridge between computers and communication.*
- *The demand for data communications is growing every day, and services like DSL and optic cable modems contribute to the growth of these infrastructures.*
- *Besides, in this new century, applications make lives easier and make communication faster and more efficient by taking advantage of the demand for more electronic methods of communications.*
- *Two, accuracy; nowadays, data isn't being very accurate and sometimes it's altered in its technical transmission.*
- *There is also the computer network; it shares different resources and is an interconnection of various computer systems that aren't in the same local place.*





Learning Points

Simple Present vs Present Continuous in Negative and Questions.

The **simple present tense** is used to express routines, habits, feeling and actions in general. On the other hand, **present continuous** expresses actions that are happening at the moment.

We use the **simple present** when we want to talk about habits or everyday activities.

The grammatical structure for negative sentences and questions is:

Negative statements:

Subject + Do/Does + Not+ verb + Complement

Questions:

Do/ Does + Subject + Verb + Complement?

*Remember that the auxiliary (Do or Does) depends on the subject you are using in the sentence or question.

*Don't forget that you can use verb to be without auxiliaries in negative sentences or questions.

*Example: The applications of data communications **aren't** the core for networks.*

Are the applications of data communications the core for networks?

We use the **present continuous** to talk about an action that is happening at the moment.

The grammatical structure is:

Negative statements:

Subject + Verb Be (is/are/am) + Not+ Verb + Complement

Questions:

Verb Be (is/are/am) + Subject + Verb + Complement?

Examples

- *I don't know much about applications of data communications. What do you know about it?*
- *It isn't possible to work without data communications.*
- *The demand for data communications is growing every day, and services like DSL and optic cable modems contribute to the growth of these infrastructures.*
- *Besides, in this new century, applications make lives easier and make communication faster and more efficient by taking advantage of the demand for more electronic methods of communications.*





Learning Points

List of Programming and Computer Science Words

This is a **list of programming and computer science words**.

- Callback: a call someone returns
- Call-by-Reference: evaluation strategy in computer programming where a function can access to a variable by using its reference.
- Call-by-Value: evaluation strategy in computer programming that describes changes not allowed on a new variable based on the same old variable.
- Careware/Charityware: software that benefits charity.
- Casting Variable: different variables to create in the computer.

Examples

- *Servers like World Wide Web pages, electronic mail, online databases, care ware and multimedia content.*
- *It isn't possible to work without data communications even when casting variables.*
- *Besides, in this new century, applications make lives easier and make communication faster and more efficient by taking advantage of the demand for more electronic methods of communications or by using strategies of evaluation like "call-by-reference" or "call-by-value" in computers programming.*

